EXPERIMENT 13

LIMITATIONS OF LOW-CODE DEVELOPMENT PLATFORMS USING ZOHO CREATOR

Aim

To study and analyze the limitations of low-code development platforms using Zoho Creator—focusing on customization, performance, scalability, integration, security, cost, and platform dependency—and to identify possible mitigation solutions for the **Application Form Management System**.

Procedure

1. Study the capabilities of low-code platforms and the scope of enterprise web applications they build.
2. Examine the **Application Form Management System**, reviewing forms, reports, workflows, and field constraints.
3. Identify areas where layout and component customization are restricted.
4. Evaluate performance, storage, and API integration constraints as application volume increases.
5. Analyze automation limits regarding complex calculations, third-party integrations, and notifications.
6. Review subscription tier thresholds and premium feature limitations.
7. Evaluate security governance, offline access dependencies, and advanced AI extensions.
8. Record observed limitations and document corresponding engineering solutions for each.

Analysis of Low-Code Platform Limitations

1. Limited Customization

Predefined layouts restrict full control over the applicant portal UI.
Solution: Use HTML/JS custom pages and widget embedding.

2. Performance Bottlenecks

Slower processing with large record sets and file uploads.
Solution: Optimize Deluge scripts and use criteria-based fetching.

3. Scalability Boundaries

High submission volumes increase API and storage requirements.
Solution: Monitor usage tiers and offload assets to external cloud storage.

4. Integration Constraints

Connecting external payment gateways requires custom APIs.
Solution: Utilize Deluge REST `invokeUrl` webhooks and REST endpoints.

5. Vendor Lock-In

Deluge code and configurations are platform-dependent.
Solution: Maintain complete schema documentation and periodic data backups.

6. Subscription & Cost Limits

Feature access and user counts are restricted by pricing plans.
Solution: Select plan tiers matching admission cycle demands.

7. Security & Compliance

Risk of exposing applicant documents via misconfigured permissions.
Solution: Enforce strict RBAC, field-level rules, and audit logs.

8. Limited Offline Access

Cloud portal relies heavily on stable network connectivity.
Solution: Establish offline paper-form fallback workflows.

Summary of Limitations Observed in This Project

Limitation Category	Observed Example in Application Form	Mitigation / Solution
Customization	Standard form layout limits custom branding in candidate portal	Custom Deluge HTML pages and widget components
Performance	Large document upload loads and high submission volumes	Criteria-based record fetching and script optimization
Scalability	High peak traffic during deadline days	Capacity monitoring and plan tier upgrades
Integration	Connecting external payment gateways & verification services	REST APIs, Deluge webhooks, and secure connectors
Vendor Lock-in	Deluge logic is non-portable to standard JS/Python runtimes	Comprehensive architecture docs and data exports
Security	Protecting applicant certificates and fee transaction receipts	Role-based access control, field restrictions, audit trails
Offline Access	Portal unavailable in remote locations without internet access	Provisional offline paper entry and batch sync
Advanced AI Tech	Automated OCR document verification for marksheets	Integrating external AI APIs with security controls

Result

The limitations of low-code development platforms were systematically studied using the **Application Form Management System**, and practical engineering solutions were identified and documented for each constraint.

Conclusion

Zoho Creator accelerates application development through pre-built forms, reports, and workflow automation. However, factors such as customization boundaries, script performance, integration mechanisms, vendor lock-in, cost structures, and security governance must be carefully evaluated during system design and deployment.